MILITARY SPECIFICATION

CAP-PLUG, CAP, AND PLUG, PROTECTIVE, PLASTIC

DUST AND MOISTURE SEAL

1. SCOPE

1.1 Scope. This specification covers plastic dust and moisture seals for use in preservation, packaging, and packing for shipment and storage.

1.2 Classification. Seals shall be furnished in the following types, forms, classes, styles and in one or more of the sizes listed in tables I through XV, figures 1 through 9 (see 6.2):

Type I - Cap-Plug - For general purposes.
Type II - Plug.
  Form A - Unthreaded.
    Class 1 - For tubing.
    Class 2 - For electrical connectors.
  Form B - Threaded.
    Class 1 - National pipe standard.
      Style a - Plain.
      Style b - Short head.
      Style c - Square head.
      Style d - Hexagon socket.
      Style e - Special shoulder.
    Class 2 - Machine screw.
      Style a - Shoulder
      Style b - Bevel seat.
      Style c - For AN-JIC fittings (aircraft and hydraulic flared and boss fittings.)
      Style d - For SAE fittings (refrigeration, automotive, etc., flared fittings.)
      Style e - For inverted flared fittings.
    Class 3 - Metric.
Type III - Cap.
  Form A - Unthreaded.
    Class 1 - For tubing.
    Class 2 - For electrical connectors.
  Form B - Threaded.
    Class 1 - National pipe standard.
    Class 2 - Machine screw.

FSC 5340

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2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L-M-505 - Molding Plastic, Cellulose Acetate.
L-P-590 - Plastic Compounds, Molding and Extrusion, Polyethylene.
PPP-B-636- Boxes, Fiber.
PPP-T-60 - Tape; Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing.

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MIL-P-20689 - Plastic, Plastisol; Molding, Extruding, Casting and Dipping Compound.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

MS25177 - Cap, Plastic, Dust, AN Electric Connector, Plug.

MS25178 - Cap, Plastic, Dust, AN Electric Connector, Receptacle.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

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3. REQUIREMENTS

3.1 Materials. All materials shall conform to specifications forming a part of this specification.

3.1.1 Type I. Type I seals shall be made of plastic compound, conforming to type I, grade 2 or 3 of Specification L-P-590 or plastisol plastic conforming to type I (class optional) of Specification MIL-P-20689.

3.1.2 Type II and III.

3.1.2.1 Form A. Form A seals shall be made of plastic compound conforming to type I, grade 2 or 3 of Specification L-P-590 or plastisol plastic conforming to type I (class optional) of Specification MIL-P-20689.

3.1.2.2 Form B. Form B seals shall be made of molding plastic conforming to type I (class optional) of Specification L-M-505 or plastic compound conforming to type I (class optional) of Specification L-P-590.

3.2 Manufacture. Seals shall be manufactured as specified in tables I through XV and figures 1 through 9 and in accordance with the type, form, class, style, and size specified in the contract or order. Unless otherwise specified the color of the seals shall be the manufacturer's standard (see 6.2). Threaded seals shall conform to National Bureau of Standards Handbook H28 and shall be suitable for use in mating parts manufactured to class 1, 2, and 3 fits. Where thread length or form specified in tables or figures is not in accordance with the manufacturer's commercial design, the manufacturer's design shall be accepted provided the seal has a minimum of three effective threads with a minimum thread engagement of 50 percent. When plugs are not threaded to the shoulder, or caps to the inside depth, an undercut shall be provided to allow proper seating. The number of ribs on the seals shall be in accordance with the commercial design for the size and type.

3.3 Electrical connectors.

3.3.1 Type II, form A, class 2. Sizes for these caps shall be as specified on MS25177.

3.3.2 Type III, form A, class 2. Sizes for these caps shall be as specified on MS25178.

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3.4 Tolerances. The following tolerances shall apply to all dimensions except threads and those specified on MS standards:

Thickness	± 0.005 inch (fractional and decimal).
Other dimensions - Up to 1-inch	± 1/64 (fractional) ± 0.010 inch (decimal)
1 inch to 2 inches	± 1/64 (fractional) ± 0.020 inch (decimal)
2 inches to 3 inches	± 1/32 (fractional) ± 0.030 inch (decimal)
Over 3 inches	± 1/32 (fractional) ± 0.040 inch (decimal)

3.5 Workmanship. The seals shall be free of contamination (oil, grease or foreign matter imbedded or on surfaces), cracks, breaks, chips or any other damage or deformation which would impair their use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all material and components are in accordance with 3.1, 3.2 and section 5 (see 6.3).

4.3 Alternate inspection (including testing) procedures. The supplier may utilize any alternate inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of the specification will govern.

4.4 Inspection provisions.

4.4.1 Regular production lot. A regular production lot shall consist of the identical seals produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions.

4.4.2 Sampling non-destructive examination or test. Sampling shall be conducted in accordance with Standard MIL-STD-105. The level and degree of inspection will be as specified in the contract.

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Table I - Sizes for type I

Size	A Plug minor diam. (in.)	B Plug major diam. (in.)	C Cap minor diam. (in.)	D Cap major diam. (in.)	E Height (in.)	F Flange diam (in.)
1	0.116	0.182	0.053	0.120	3/8	5/16
2	.173	.240	.111	.177	3/8	3/8
3	.236	.302	.174	.240	3/8	7/16
4	.299	.365	.235	.302	3/8	1/2
5	.330	.396	.267	.333	3/8	17/32
6	.361	.427	.298	.365	3/8	9/16
7	.390	.456	.327	.394	3/8	19/32
8	.419	.485	.356	.422	3/8	5/8
9	.447	.513	.385	.451	3/8	21/32
10	.471	.547	.409	.485	7/16	11/16
11	.532	.608	.468	.545	7/16	3/4
12	.550	.631	.487	.568	15/32	3/4
13	.580	.667	.518	.605	1/2	13/16
14	.640	.727	.578	.664	1/2	7/8
15	.675	.755	.613	.693	1/2	7/8
16	.690	.788	.628	.726	9/16	15/16
17	.725	.823	.663	.761	9/16	31/32
18	.750	.848	.687	.785	9/16	1
19	.802	.911	.740	.849	5/8	1-1/16
20	.866	.975	.803	.912	5/8	1-1/8
21	.915	1.040	.835	.960	11/16	1-5/32
22	.964	1.090	.901	1.027	3/4	1-1/4
23	1.094	1.215	1.031	1.152	3/4	1-3/8
24	1.175	1.284	1.113	1.222	5/8	1-7/16
25	1.224	1.355	1.161	1.292	3/4	1-1/2
26	1.312	1.465	1.249	1.402	7/8	1-5/8
27	1.447	1.600	1.384	1.537	7/8	1-3/4
28	1.560	1.682	1.498	1.620	3/4	1-7/8
29	1.750	1.880	1.688	1.818	3/4	2
30	1.780	1.950	1.718	1.888	1	2-1/8
31	1.900	2.076	1.837	2.013	1	2-1/4
32	1.290	1.355	1.227	1.292	3/8	1-1/2
33	1.513	1.600	1.451	1.537	1/2	1-3/4
34	1.800	1.824	1.738	1.762	3/8	2
35	1.787	1.963	1.725	1.901	1	2-3/16
36	1.844	2.000	1.781	1.938	1	2-1/8
37	1.975	2.132	1.913	2.070	1	2-1/4
38	2.015	2.147	1.953	2.085	3/4	2-5/16
39	2.067	2.243	2.005	2.181	1	2-7/16
40	2.136	2.312	2.074	2.250	1	2-9/16

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Table I - Sizes for type I (Continued)

Size	A Plug minor diam. (in.)	B Plug major diam. (in.)	C Cap minor diam. (in.)	D Cap major diam. (in.)	E Height (in.)	F Flange diam. (in.)
41	2.230	2.260	2.168	2.198	1/2	2-1/2
42	2.125	2.406	2.063	2.344	1	2-17/32
43	2.298	2.464	2.235	2.401	1	2-5/8
44	2.395	2.602	2.333	2.540	1-1/8	2-13/16
45	2.480	2.644	2.418	2.582	5/8	2-25/32
46	2.307	2.650	2.245	2.588	1-1/4	2-25/32
47	2.457	2.750	2.395	2.688	1	2-7/8
48	2.641	2.790	2.579	2.728	3/4	2-15/16
49	2.720	2.990	2.658	2.922	1	3-1/8
50	2.849	2.980	2.787	2.918	3/4	3-1/4
51	2.995	3.158	2.915	3.078	3/4	3-3/8
52	2.982	3.245	2.912	3.175	1	3-3/8
53	3.090	3.370	3.020	3.300	1	3-1/2
54	3.325	3.457	3.245	3.377	3/4	3-5/8
55	3.225	3.562	3.155	3.492	1-1/8	3-11/16

Table II - Sizes for type II, form A, class 1

Size	To fit tube I.D. (in.)	A Plug major diam. (in.)	B Plug minor diam. (in.)	C Length (in.)	D Flange diam. (in.)	E Core hole diam. (in.)
1	3/16	0.195	0.182	19/32	5/16	3/32
2	1/4	.260	.245	21/32	3/8	1/8
3	5/16	.322	.307	21/32	7/16	3/16
4	3/8	.385	.370	21/32	1/2	1/4
5	7/16	.447	.432	21/32	9/16	5/16
6	1/2	.510	.495	21/32	5/8	3/8
7	9/16	.572	.557	21/32	11/16	7/16
8	5/8	.635	.620	21/32	3/4	1/2
9	11/16	.697	.682	21/32	13/16	9/16
10	3/4	.760	.745	21/32	7/8	5/8
11	13/16	.822	.807	21/32	15/16	11/16
12	7/8	.885	.870	21/32	1	3/4
13	15/16	.947	.932	21/32	1-1/16	13/16
14	1	1.010	.995	21/32	1-1/8	7/8

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Table III - Sizes for type II, form B, class 1, style a

Size	A Thread size, NPT (in.)	B Thread length (in.)	C Head height (in.)	D Head diam. (in.)
1	1/8	3/8	1/4	5/16
2	1/4	3/8	1/4	3/8
3	3/8	3/8	1/4	1/2
4	1/2	3/8	1/4	9/16
5	3/4	3/8	1/4	13/16
6	1	3/8	1/4	1
7	1-1/4	3/8	1/4	1
8	1-1/4	1/2	1/4	1
9	1-1/2	1/2	1/4	1
10	2	1/2	1/4	1

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Table IV - Sizes for type II, form B, class 1, style b

Size	A Thread size, NPT (in.)	B Thread length (in.)	C Head height (in.)	D Head diam. (in.)
1	1/8	1/4	1/16	5/16
2	1/4	3/8	1/16	3/8

Table V - Sizes for type II, form B, class 1, style c

Size	A Thread size, NPT (in.)	B Thread length (in.)	C Head height (in.)	D Head square (in.)
1	1/8	5/16	1/4	15/64
2	1/4	5/16	1/4	3/8
3	3/8	5/16	1/4	7/16
4	1/2	5/16	1/4	1/2

Table VI - Sizes for type II, form B, class 1, style d

Size	A Thread size, NPT (in.)	B Major diam. Iin.)	C Overall height (in.)	D Hex hole depth (in.)	E Hex hole size (in.)
1	1/4	0.540	1/2	1/4	1/4
2	3/8	.675	9/16	3/8	5/16
3	1/2	.840	9/16	3/8	3/8
4	3/4	1.050	5/8	3/8	9/16
5	1	1.315	3/4	1/2	5/8
6	1-1/4	1.660	3/4	5/8	3/4
7	1-1/2	1.900	3/4	5/8	3/4

Table VII - Sizes for type II, form B, class 1, style e

Size	A Thread size, NPT (in.)	B Thread length (in.)	C Shoulder height (in.)	D Shoulder diam. (in.)
1	1/2	7/16	3/8	1-1/4
2	3/4	7/16	3/8	1-5/8
3	1	7/16	3/8	2

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Table VIII - Sizes for type II, form B, class 2, style a

Size	A Thread size	B Thread length (in.)	C Head height (in.)	D Head diam. (in.)	E Shoulder thickness (in.)	F Shoulder diam. (in.)
1	1/4 - 28	5/16	9/32	3/8	0.065	1/2
2	5/16 - 20	5/16	9/32	3/8	.065	1/2
3	5/16 - 24	5/16	9/32	3/8	.065	1/2
4	3/8 - 24	5/16	9/32	3/8	.065	5/8
5	7/16 - 14	5/16	9/32	1/2	.075	11/16
6	7/16 - 20	5/16	9/32	1/2	.075	11/16
7	7/16 - 24	5/16	9/32	1/2	.075	11/16
8	1/2 - 13	3/8	9/32	1/2	.085	3/4
9	1/2 - 20	3/8	9/32	1/2	.085	3/4
10	9/16 - 12	7/32	9/32	9/16	.093	13/16
11	9/16 - 18	3/8	9/32	9/16	.093	13/16
12	9/16 - 20	3/8	9/32	9/16	.093	13/16
13	5/8 - 18	3/8	9/32	9/16	.093	7/8
14	5/8 - 24	3/8	9/32	9/16	.093	7/8
15	5/8 - 27	3/8	9/32	9/16	.093	7/8
16	11/16 - 16	3/8	9/32	9/16	.093	15/16
17	11/16 - 18	9/32	9/32	9/16	.093	15/16
18	3/4 - 16	3/8	1/4	25/32	.093	1
19	3/4 - 18	3/8	1/4	25/32	.093	1
20	3/4 - 20	3/8	1/4	25/32	.093	1
21	3/4 - 24	1/4	1/4	25/32	.093	1
22	7/8 - 14	3/8	1/4	25/32	.093	1-1/16
23	7/8 - 18	3/8	1/4	25/32	.093	1-1/16
24	7/8 - 20	3/8	1/4	25/32	.093	1-1/16
25	29/32 - 14	3/8	1/4	25/32	.093	1-1/8
26	1 - 12	3/8	1/4	25/32	.093	1-3/16
27	1 - 14	3/8	1/4	25/32	.093	1-3/16
28	1 - 18	3/8	1/4	25/32	.093	1-1/4
29	1 - 20	19/32	9/32	25/32	.093	1-1/4
30	1-1/16 - 12	3/8	1/4	25/32	.093	1-1/4
31	1-1/16 - 14	3/8	1/4	25/32	.093	1-1/4
32	1-1/8 - 12	3/8	1/4	25/32	.093	1-5/16
33	1-1/4 - 12	3/8	1/4	1	.093	1-7/16
34	1-1/4 - 18	3/8	1/4	1	.093	1-7/16
35	1-5/16 - 12	3/8	1/4	1	.093	1-1/2
36	1-5/16 - 14	3/8	1/4	1	.093	1-1/2
37	1-3/8 - 12	3/8	1/4	1	.093	1-9/16
38	1-3/8 - 18	3/8	1/4	1	.093	1-9/16
39	1-1/2 - 20	3/8	1/4	1	.093	1-11/16
40	1-5/8 - 12	1/2	1/4	1	.093	1-7/8
41	1-5/8 - 14	1/2	1/4	1	.093	1-7/8
42	1-7/8 - 12	1/2	1/4	1	.093	2-1/8
43	1-7/8 - 14	1/2	1/4	1	.093	2-1/8
44	2 - 16	1/2	1/4	1	.093	2-1/4

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Table IX - Sizes for type II, form B, class 2, style b

Size	A Thread size	B Thread length (in.)	C Head height (in.)	D Overall length under shoulder (in.)	E Pilot diam. (in.)	F Shoulder diam. (in.)	G Shoulder thickness (in.)
1	3/8 - 24	3/16	21/64	27/64	5/16	9/16	5/32
2	7/16 - 20	3/16	23/64	7/16	23/64	11/16	5/32
3	1/2 - 20	1/4	23/64	29/64	27/64	3/4	5/32
4	9/16 - 18	1/4	3/8	15/32	31/64	13/16	5/32
5	5/8 - 18	5/16	7/16	17/32	33/64	7/8	5/32
6	3/4 - 16	1/4	27/64	17/32	5/8	1	5/32
7	3/4 - 16	5/16	1/2	5/8	5/8	1	5/32
8	7/8 - 14	1/4	33/64	41/64	3/4	1-1/8	5/32
9	7/8 - 14	5/16	21/32	25/32	3/4	1-1/8	5/32
10	7/8 - 14	3/8	19/32	23/32	3/4	1-1/8	5/32
11	7/8 - 14	3/8	3/4	1	3/4	1-1/8	5/32
12	1-1/16 - 12	1/4	9/16	21/32	15/16	1-3/8	5/32
13	1-1/16 - 12	1/2	17/32	21/32	15/16	1-3/8	5/32
14	1-1/16 - 14	3/8	27/32	1	15/16	1-3/8	5/32
15	1-5/16 - 12	1/4	19/32	23/32	1-3/16	1-39/64	5/32
16	1-5/16 - 12	1/2	9/16	11/16	1-3/16	1-39/64	5/32
17	1-5/8 - 12	5/16	43/64	49/64	1-7/16	2-1/8	9/64
18	1-5/8 - 12	1/2	43/64	49/64	1-7/16	2-1/8	9/64
19	1-7/8 - 12	5/16	3/4	27/32	1-11/16	2-3/8	9/64

Table X - Sizes for type II, form B, class 2, style c and d

Size	A Thread size	B Flange thickness (in.)	C Body length (in.)	D Body diameter (in.)	E Flange diameter (in.)	F Chamfer (in.)	Tubing size OD
1	5/16 - 24	3/16	7/16	0.270	7/16	1/8	1/8
2	3/8 - 24	3/16	1/2	0.332	1/2	1/8	3/16
3	7/16 - 20	3/16	17/32	0.387	5/8	1/8	1/4
4	1/2 - 20	1/4	17/32	0.449	11/16	1/8	5/16
*5	9/16 - 18	1/4	1/2	0.506	13/16	1/8	3/8
**6	5/8 - 18	1/4	19/32	0.569	13/16	3/16	3/8
7	3/4 - 16	1/4	3/4	0.686	15/16	1/4	1/2
8	7/8 - 14	1/4	7/8	0.802	1-1/16	1/4	5/8
*9	1-1/16 - 12	1/4	7/8	0.977	1-1/4	1/4	3/4
**10	1-1/16 - 14	1/4	1-1/8	0.978	1-3/8	1/4	3/4
*11	1-5/16 - 12	1/4	15/16	1.227	1-1/2	1/4	1
*12	1-5/8 - 12	1/4	15/16	1.540	2	1/4	1-1/4
*13	1-7/8 - 12	5/16	1-1/4	1.790	2-3/16	3/8	1-1/2
*14	2-1/2 - 12	1/4	1-5/16	2.415	2-7/8	3/8	2

*Applicable only to type II, form B, class 2, style c.
**Applicable only to type II, form B, class 2, style d.

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Table XI - Sizes for type II, form B, class 2, style e

Size	A Thread size	B Flange thickness (in.)	C body length (in.)	D Flange diameter (in.)	E Edge to first thread (in.)	Tubing size OD
1	5/16 - 28	3/16	7/32	7/16	1/16	1/8
2	3/8 - 24	3/16	1/4	1/2	1/16	3/16
3	7/16 - 24	3/16	17/64	5/8	1/16	1/4
4	1/2 - 20	1/4	9/32	11/16	1/16	5/16
5	5/8 - 18	1/4	5/16	13/16	1/16	3/8
6	3/4 - 18	1/4	3/8	7/8	1/16	1/2
7	7/8 - 18	1/4	3/8	1-1/16	1/16	5/8
8	1-1/16 - 16	1/4	7/16	1-3/16	1/16	3/4

Table XII - Sizes for type II, form B, class 3

Size	A Thread diam. (mm.)	A Thread pitch	B Thread length (in.)	C Head height (in.)	D Hex. across flats (in.)	E Shoulder thickness (in.)	F Shoulder diam. (in.)
1	14	1.5	7/16	15/64	0.615	0.080	3/4
2	18	1.5	7/16	15/64	0.665	0.080	15/16

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Table XIII - Sizes for type III, form A, class 1

Size	To fit tube having O.D. (in.)	A Inside diameter (in.)	B Inside length (in.)
1	3/32	3/32	1/2
2	1/8	1/8	19/32
3	5/32	5/32	19/32
4	3/16	3/16	19/32
5	1/4	1/4	19/32
6	5/16	5/16	19/32
7	3/8	3/8	19/32
8	7/16	7/16	19/32
9	15/32	15/32	19/32
10	1/2	1/2	19/32
11	17/32	17/32	5/8
12	9/16	9/16	5/8
13	5/8	5/8	5/8
14	11/16	11/16	5/8
15	3/4	3/4	5/8
16	13/16	13/16	5/8
17	7/8	7/8	5/8
18	1	1	5/8
19	1-1/16	1-1/16	5/8
20	1-1/8	1-1/8	5/8
21	1-3/16	1-3/16	5/8
22	1-1/4	1-1/4	5/8
23	1-1/4	1-1/4	7/8
24	1-5/16	1-5/16	5/8
25	1-3/8	1-3/8	5/8
26	1-3/8	1-3/8	7/8
27	1-7/16	1-7/16	5/8
28	1-1/2	1-1/2	5/8
29	1-1/2	1-1/2	7/8
30	1-5/8	1-5/8	5/8
31	1-3/4	1-3/4	5/8
32	1-3/4	1-3/4	7/8
33	1-7/8	1-7/8	5/8
34	1-7/8	1-7/8	7/8
35	2	2	5/8
36	2	2	1
37	2-1/8	2-1/8	1
38	2-1/4	2-1/4	5/8
39	2-1/4	2-1/4	1
40	2-3/8	2-3/8	1
41	2-1/2	2-1/2	5/8
42	2-1/2	2-1/2	1
43	2-5/8	2-5/8	1
44	2-3/4	2-3/4	5/8
45	2-3/4	2-3/4	1
46	2-7/8	2-7/8	1
47	3	3	5/8
48	3	3	1

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Table XIV - Sizes for type III, form B, class 1

Size	A Thread size, NPT (in.)	B Outside diam. (in.)	C Inside length (in.)
1	1/8	17/32	3/8
2	1/4	11/16	3/8
3	3/8	27/32	7/16
4	1/2	1-1/16	7/16
5	3/4	1-1/4	1/2
6	1	1-1/2	1/2
7	1-1/4	1-13/16	1/2
8	1-1/2	2-1/16	1/2

Table XV - Sizes for type III, form B, class 2

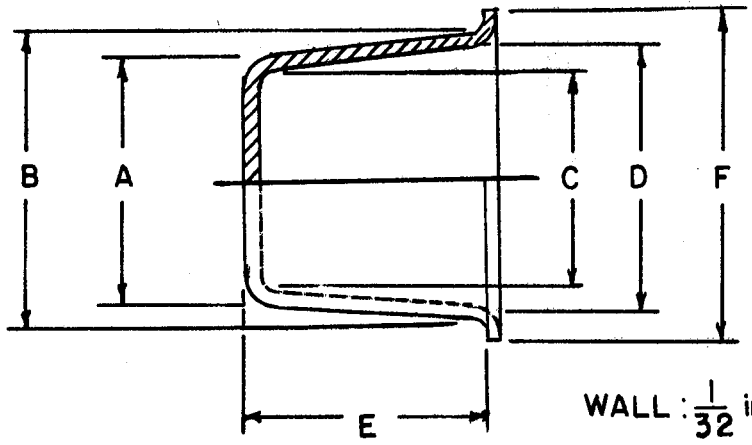
Size	Thread size	B Outside diam. (in.)	C Inside length (in.)
1	1/4 - 28	3/8	5/16
2	5/16 - 24	7/16	3/8
3	3/8 - 24	1/2	3/8
4	3/8 - 32	1/2	3/8
5	7/16 - 20	19/32	3/8
6	1/2 - 20	11/16	3/8
7	1/2 - 28	11/16	7/16
8	9/16 - 12	3/4	3/8
9	9/16 - 18	3/4	3/8
10	9/16 - 18	3/4	9/16
11	9/16 - 20	3/4	3/8
12	5/8 - 18	13/16	7/16
13	5/8 - 18	13/16	9/16
14	5/8 - 20	13/16	3/4
15	5/8 - 24	13/16	7/16
16	5/8 - 27	13/16	7/16
17	11/16 - 16	7/8	7/8
18	11/16 - 24	7/8	7/16
19	3/4 - 16	15/16	7/16
20	3/4 - 16	15/16	11/16
21	3/4 - 18	15/16	7/16
22	3/4 - 20	15/16	1/2
23	13/16 - 16	1	1/2
24	7/8 - 14	1-1/16	7/16
25	7/8 - 14	1-1/16	11/16

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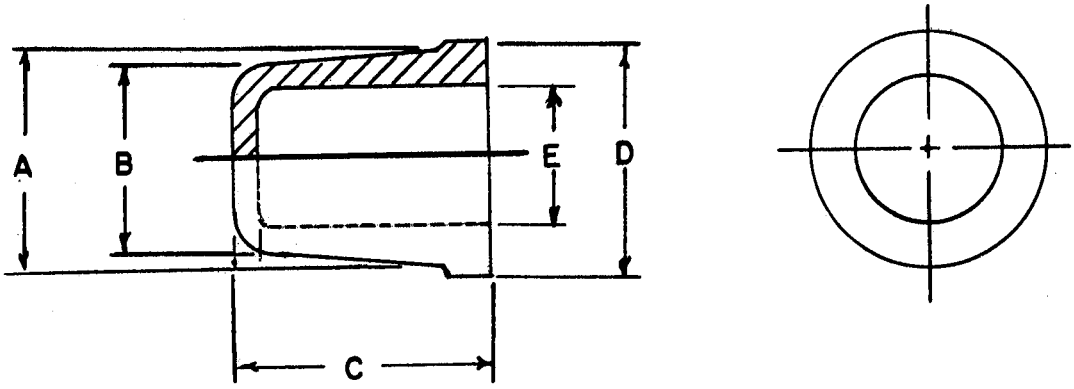
Table XV - Sizes for type III, form B, class 2 (Continued)

Size	A Thread size	B Outside diam. (in.)	C Inside length (in.)
26	7/8 - 18	1-1/16	7/16
27	7/8 - 20	1	7/16
28	29/32 - 14	1-1/16	1/2
29	15/16 - 18	1-1/16	7/16
30	1 - 14	1-5/32	1/2
31	1 - 20	1-3/16	1/2
32	1-1/16 - 12	1-1/4	1/2
33	1-1/16 - 14	1-1/4	23/32
34	1-1/16 - 16	1-1/4	23/32
35	1-1/8 - 12	1-1/4	3/4
36	1-1/8 - 18	1-9/32	1/2
37	1-3/16 - 18	1-11/32	1/2
38	1-1/4 - 12	1-1/2	1/2
39	1-1/4 - 18	1-7/16	1/2
40	1-5/16 - 12	1-1/2	9/16
41	1-5/16 - 14	1-1/2	9/16
42	1-3/8 - 6	1-9/16	5/8
43	1-3/8 - 12	1-9/16	5/8
44	1-3/8 - 18	1-9/16	1/2
45	1-1/2 - 16	1-11/16	3/4
46	1-1/2 - 18	1-11/16	9/16
47	1-5/8 - 12	1-13/16	9/16
48	1-5/8 - 16	1-13/16	1/2
49	1-3/4 - 18	1-15/16	9/16
50	1-7/8 - 12	2-1/16	9/16
51	1-7/8 - 12	2-1/16	1-5/32
52	2 - 16	2-3/16	9/16
53	2 - 18	2-3/16	9/16
54	2-1/4 - 16	2-7/16	9/16

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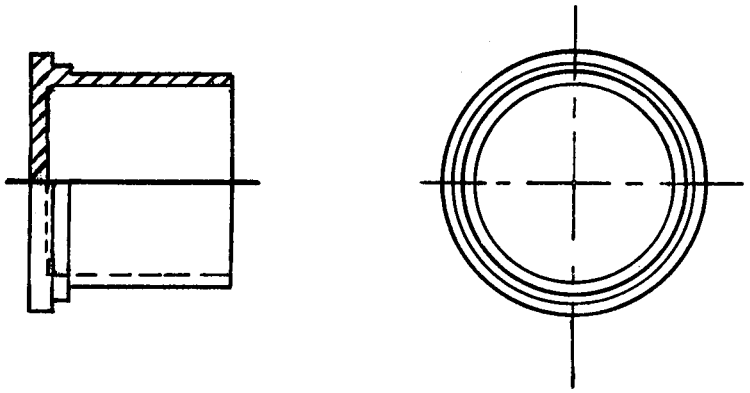
Type I
(see table I)



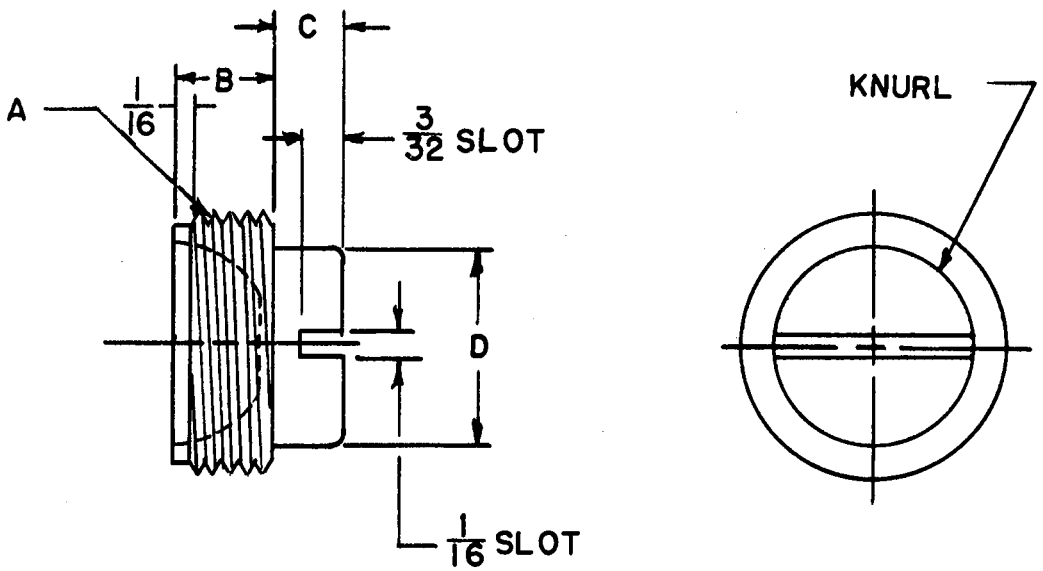
Type II, form A, class 1
(see table II)

Figure 1.- Cap - Plug and plug

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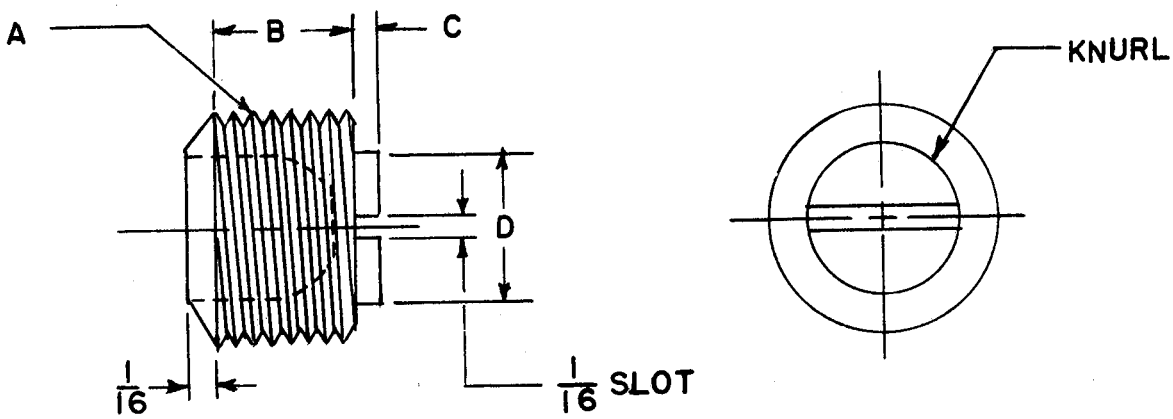
Type II, form A, class 2
(see 3.3.1)



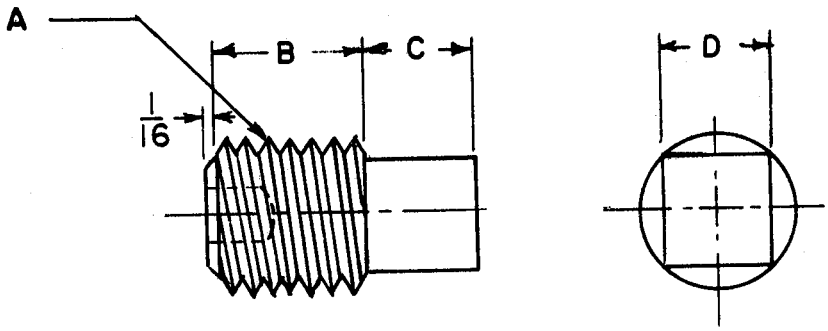
Type II, form B, class 1, style a
(see table III)

Figure 2. - Plugs

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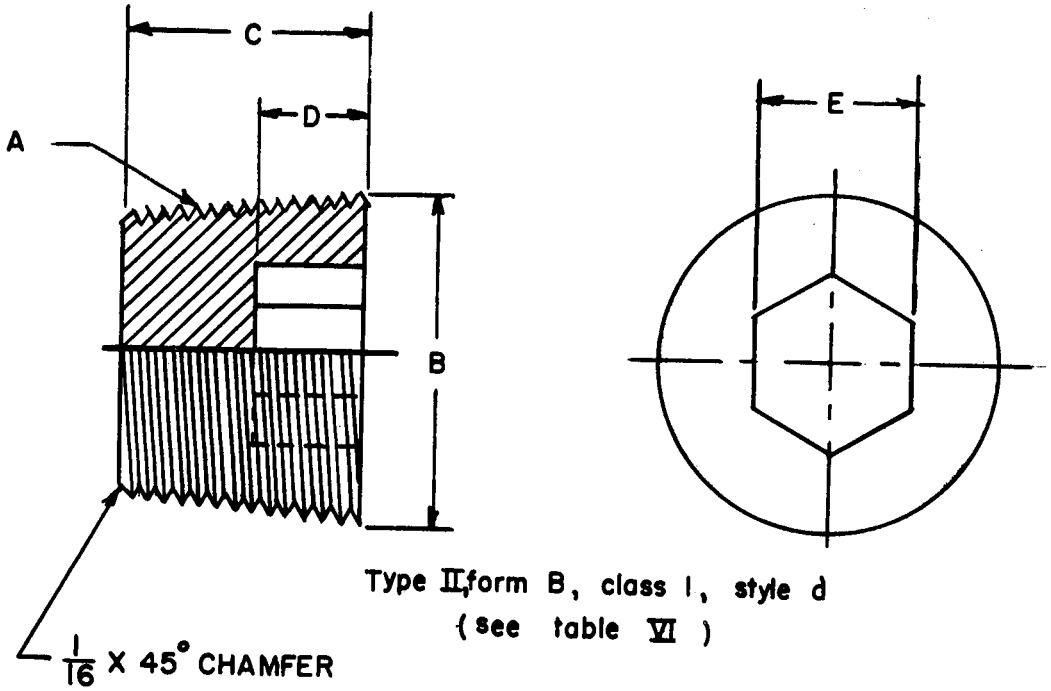
Type II, form B, class 1, style b
(see table IV)



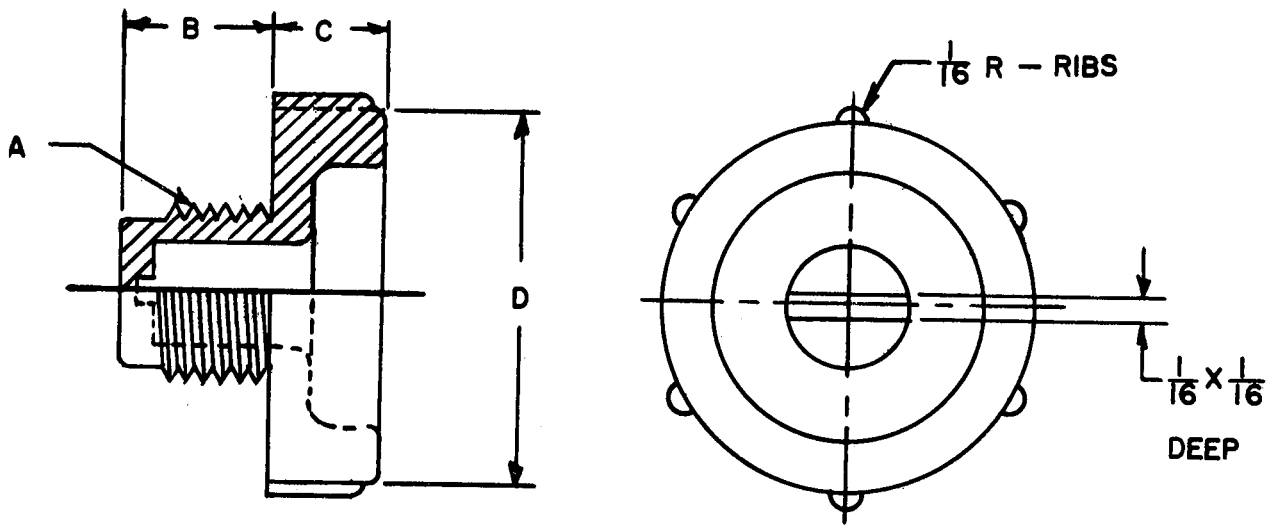
Type II, form B, class 1, style c
(see table V)

Figure 3.- Plugs

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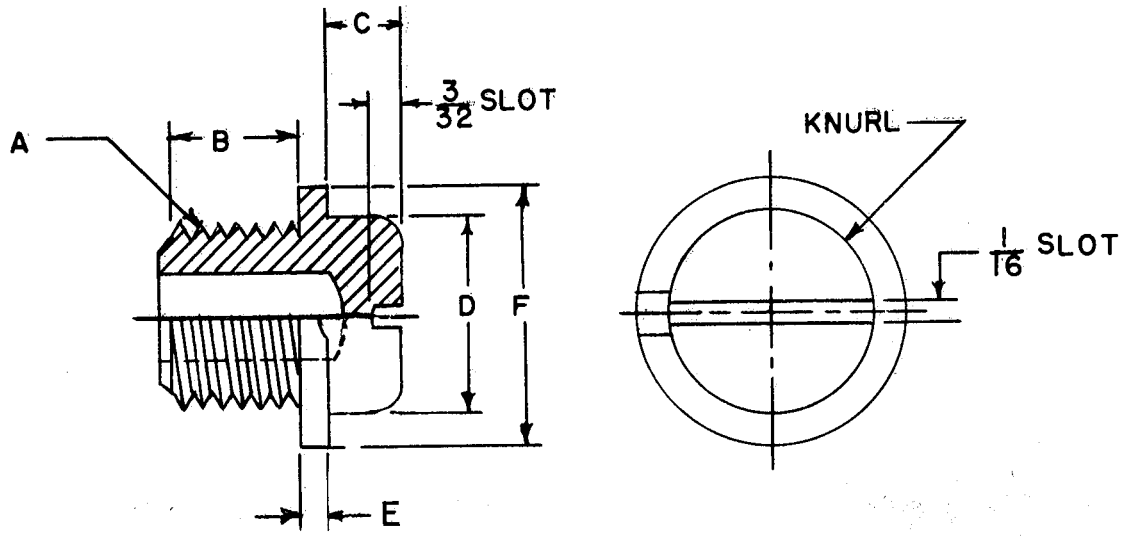
Type II, form B, class 1, style d
(see table VI)



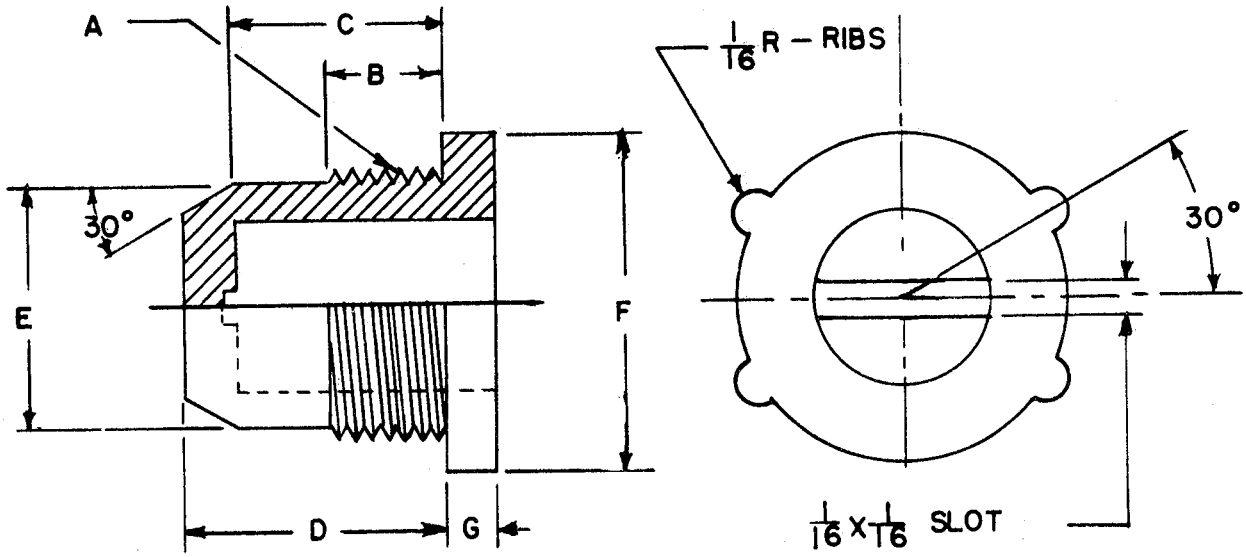
Type II, form B, class 1, style e
(see table VII)

Figure 4 - Plugs

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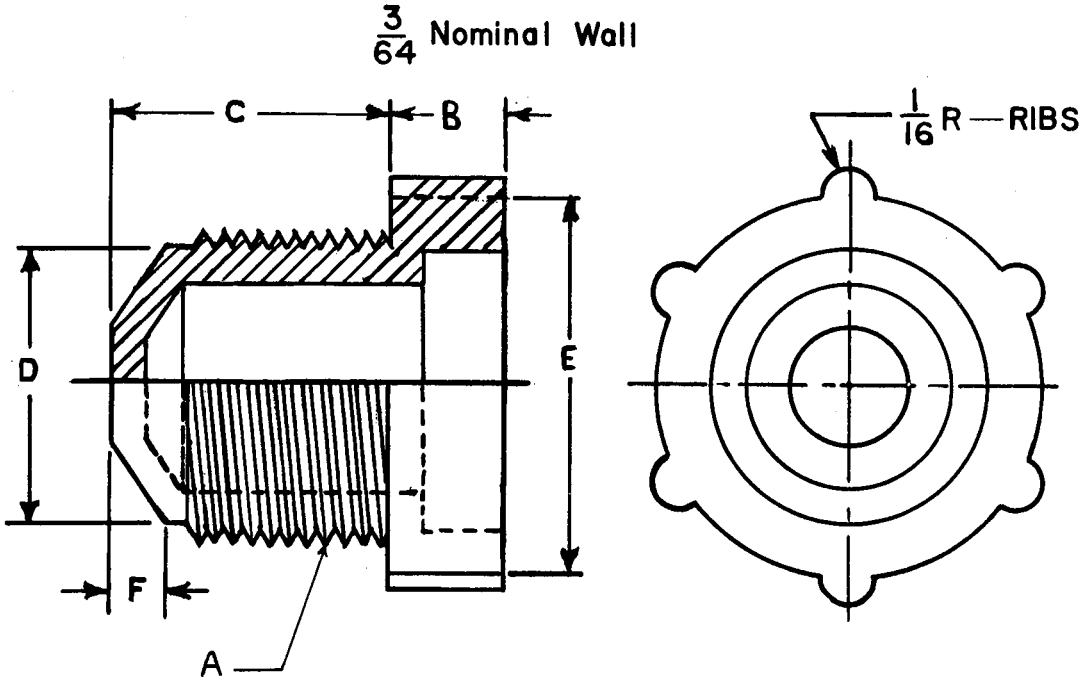
Type II, form B, class 2, style a
(see table VIII)



Type II, form B, class 2, style b
(see table IX)

Figure 5.- Plugs

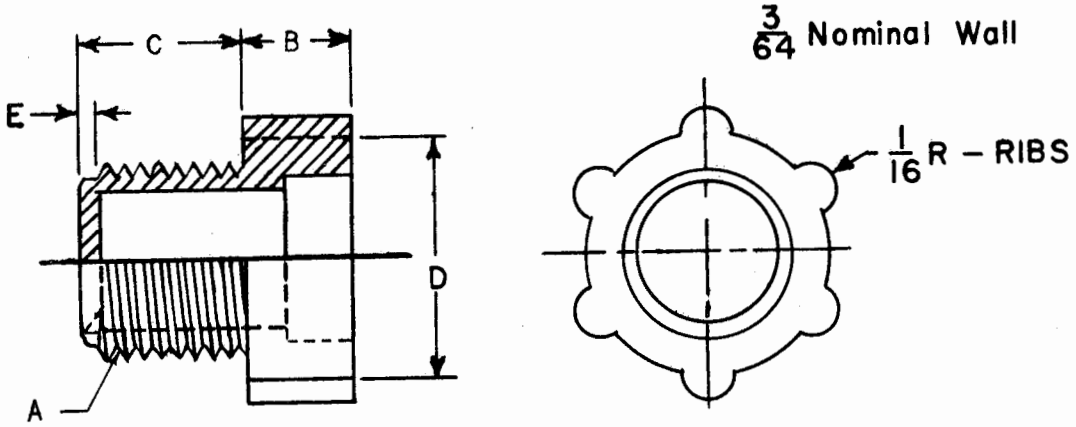
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:06Z
Check the source to verify that this is the current version before use.



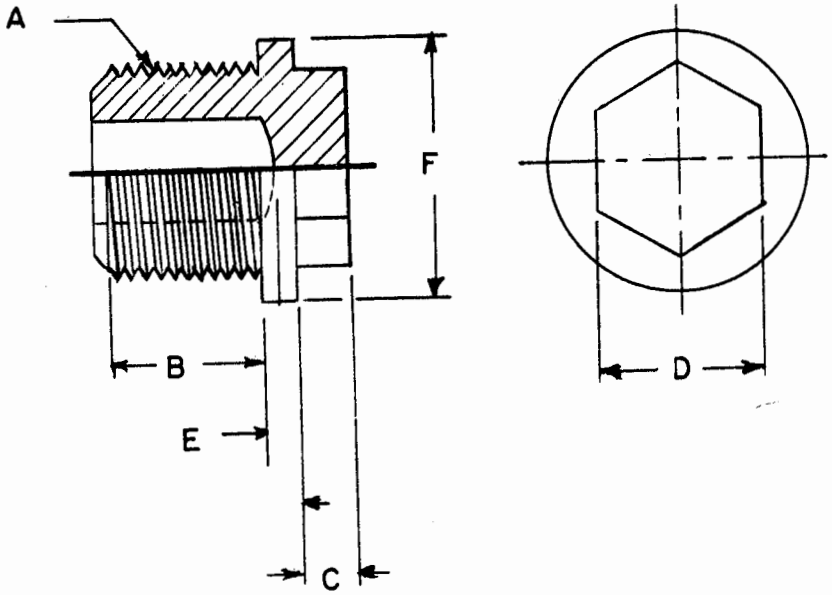
Type II, form B, class 2, style c and d
(see table X)

Figure 6.- Plug

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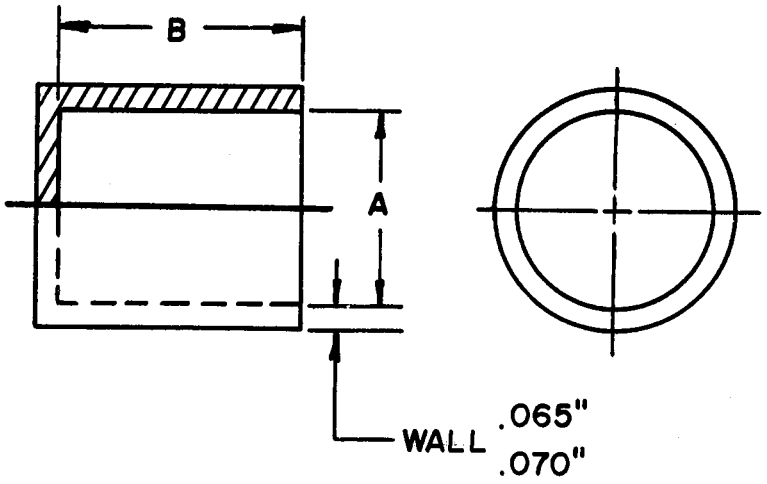
Type, II, form B, class 2, style e
(see table XI)



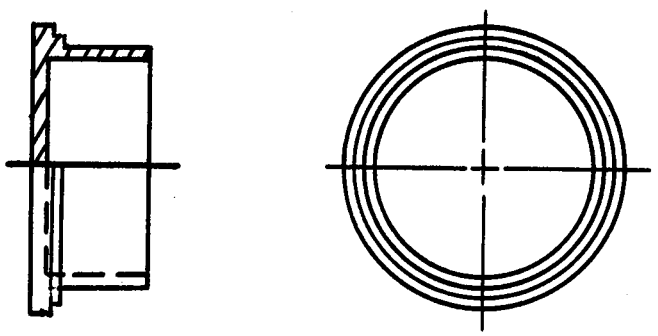
Type II, form B, class 3
(see table XII)

Figure 7.- Plugs

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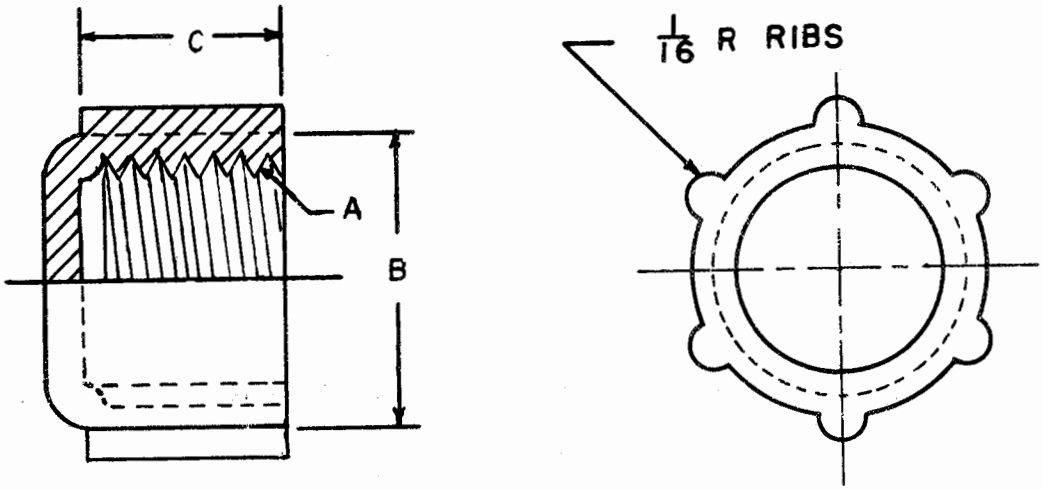
Type III, form A, class 1
(see table XIII)



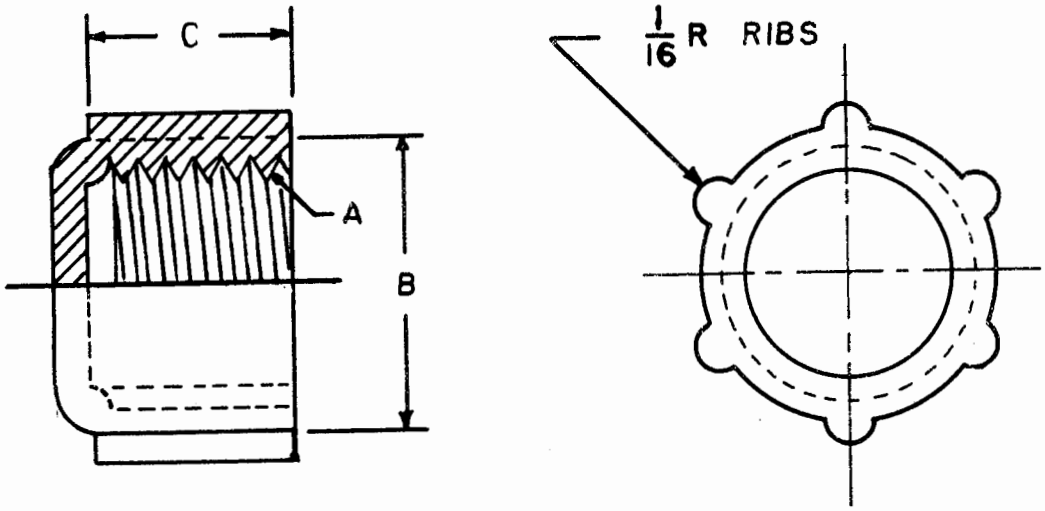
Type III, form A, class 2
(see table 3.3.2)

Figure 8. - Caps

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Type III, form B, class 1
(see table XIV)



Type III, form B, class 2
(see table XV)

Figure 9.- Caps

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4.4.3 Examination. Sample seals from the regular production lots shall be examined in accordance with the classification of defects and the AQL's as designated by this specification and with Standard MIL-STD-105.

4.4.4 Classification of defects .

4.4.4.1 Preparation for delivery (Section 5).

Major	AQL 6.5 percent defective
1. Container incorrect	
2. Container closure incorrect	
3. Container damaged	
4. Marking missing, illegible or incorrect	

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Packaging shall be in accordance with the manufacturer's commercial practice and shall provide protection for the closures packed as specified in 5.2.1 and 5.2.3.

5.2 Packing.

5.2.1 Level A. A uniform quantity of seals, packaged as specified in 5.1.3, shall be packed in a snug-fitting V3c, style RSC fiberboard box conforming to type I, class 2 of Specification PPP-B-636. Only seals of one type, form, class, style, and size shall be packed in each box. The box shall be closed in accordance with the appendix to the box specification for exterior boxes. When specified all seams including the manufacturer's joint shall be taped with pressure sensitive tape having a minimum width of 2 inches and conforming to type III, class 1 (color optional) of Specification PPP-T-60 (see 6.2). The tape shall extend over all corners and edges a minimum distance of 3 inches.

5.2.2 Level B. Not applicable.

5.2.3 Level C. Seals packaged as specified in 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order unit packages, intermediate packages, and shipping containers shall be marked in accordance with Standard MIL-STD-129.

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6. NOTES

6.1 Intended use. Cap-plugs, caps, and plugs are plastic seals that are used for covering or plugging small openings on items being prepared for storage or shipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of packing.
(c) Type, form, class, style, and size of seal required.
(d) Color when other than manufacturer's standard.
(e) Box seams to be taped.

6.3 Objective evidence. Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

6.4 Pilot lot. A pilot lot may be required to evaluate the suppliers production and quality control systems. The contracting officer will specify whether a pilot lot is required and applicable pilot lot acceptance criteria.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:

Army - Chemical Corps

Preparing activity:

Army - Chemical Corps

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